

MH1301 Discrete Mathematics

Tutorial 3

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Overview

Key skills in this tutorial:

- counting bit strings via combinations, complements, and symmetry;
- team/committee selection under constraints;
- structured strings with adjacency rules (block method, stars-and-bars viewpoint);
- coefficients in binomial and multinomial-type expansions (binomial theorem, combinatorial reasoning);
- inclusion–exclusion for unions of sets;
- binomial identities proved algebraically and combinatorially.

Question map.

- Warm-ups: Ex. 6.3.3 (permutations/combinations), Ex. 6.3.27 (bit strings with lower bounds).
- Week 4 discussion: Ex. 6.3.7, 6.3.16, 6.3.25, 6.4.2, 6.4.5, 8.5.16, 6.4.13, 6.4.20, 6.4.21.
- Optional challenge: poker-hand counting (four-of-a-kind, full house, flush, two-pair).

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Question 1 (Warm-up: Ex. 6.3.3)

Problem

Given $S = \{a, b, c, d, e, f, g\}$:

- (a) How many 5-permutations end with a ?
- (b) How many 5-combinations contain a ?

Solution

Method 1: Direct slot-filling (product rule / combination rule)

- (a) A 5-permutation is an ordered selection of 5 distinct elements from S . If it must end with a , the last position is fixed as a . The remaining 4 positions must be filled with 4 distinct letters from $S \setminus \{a\}$, which has 6 letters. Hence the count is the number of 4-permutations of 6 objects:

$${}_6P_4 = 6 \cdot 5 \cdot 4 \cdot 3 = \frac{6!}{(6-4)!} = \frac{6!}{2!} = 360.$$

360

- (b) A 5-combination is an unordered selection of 5 distinct elements from S . If the selection must contain a , then we choose the remaining 4 elements from $S \setminus \{a\}$ (which has 6 elements). Thus

$$\binom{6}{4} = 15.$$

15

Method 2: Equivalent reformulations (remove/leave-out viewpoint)

- (a) Ending with a means: choose an ordered list of 4 distinct letters from $\{b, c, d, e, f, g\}$ and then append a . Therefore the number equals the number of injective functions from $\{1, 2, 3, 4\}$ into a 6-set:

$$6 \cdot 5 \cdot 4 \cdot 3 = 360.$$

360

- (b) Choosing 5 letters that contain a is equivalent to choosing which *two* letters are *excluded* from $S \setminus \{a\}$ (since among the other 6 letters, we include exactly 4). Hence

$$\binom{6}{2} = 15.$$

15

Question 2 (Warm-up: Ex. 6.3.27)

Problem

How many bit strings of length 10 contain at least three 1's and at least three 0's?

Solution

Method 1: Count by number of 1's

Let a bit string of length 10 have exactly k ones. Then it has $10 - k$ zeros. The condition at least three 1's and at least three 0's

is equivalent to

$$k \geq 3 \quad \text{and} \quad 10 - k \geq 3 \iff 3 \leq k \leq 7.$$

For fixed k , the number of strings with exactly k ones is $\binom{10}{k}$ (choose the positions of the ones). Therefore the desired number is

$$\sum_{k=3}^7 \binom{10}{k}.$$

Compute using complements/symmetry:

$$\sum_{k=0}^{10} \binom{10}{k} = 2^{10} = 1024,$$

and by symmetry $\binom{10}{k} = \binom{10}{10-k}$, so

$$\sum_{k=3}^7 \binom{10}{k} = 2^{10} - \left(\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10} \right).$$

Now

$$\binom{10}{0} = 1, \quad \binom{10}{1} = 10, \quad \binom{10}{2} = 45, \quad \binom{10}{8} = 45, \quad \binom{10}{9} = 10, \quad \binom{10}{10} = 1,$$

so the excluded total is $1 + 10 + 45 + 45 + 10 + 1 = 112$. Hence

$$\sum_{k=3}^7 \binom{10}{k} = 1024 - 112 = 912.$$

$$\boxed{912}$$

Method 2: Complement principle (forbidden cases)

All bit strings of length 10 total $2^{10} = 1024$.

A string fails the condition iff it has at most two 1's *or* at most two 0's. Since having at most two 0's is equivalent to having at least eight 1's, the “bad” strings are exactly those with

$$k \in \{0, 1, 2, 8, 9, 10\} \quad \text{ones.}$$

Thus the number of bad strings is

$$\sum_{k \in \{0, 1, 2, 8, 9, 10\}} \binom{10}{k} = 2 \left(\binom{10}{0} + \binom{10}{1} + \binom{10}{2} \right) = 2(1 + 10 + 45) = 112,$$

and the good strings are $1024 - 112 = 912$.

$$\boxed{912}$$

Question 3 (Ex. 6.3.7)

Problem

How many bit strings of length 10 contain

- (a) exactly four 1's?
- (b) at most four 1's?
- (c) at least four 1's?
- (d) an equal number of 0's and 1's?

Solution

Method 1: Choose positions of the 1's (combinations)

A bit string is determined by the set of positions at which a 1 occurs.

- (a) Exactly four 1's: choose 4 positions among 10:

$$\binom{10}{4} = 210.$$

210

- (b) At most four 1's: sum over $k = 0, 1, 2, 3, 4$:

$$\sum_{k=0}^4 \binom{10}{k} = \binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \binom{10}{3} + \binom{10}{4}.$$

Compute:

$$1 + 10 + 45 + 120 + 210 = 386.$$

386

- (c) At least four 1's: sum over $k = 4$ to 10:

$$\sum_{k=4}^{10} \binom{10}{k} = 2^{10} - \sum_{k=0}^3 \binom{10}{k}.$$

Compute $\sum_{k=0}^3 \binom{10}{k} = 1 + 10 + 45 + 120 = 176$, hence

$$2^{10} - 176 = 1024 - 176 = 848.$$

848

- (d) Equal number of 0's and 1's: length 10 implies 5 ones and 5 zeros, so

$$\binom{10}{5} = 252.$$

252

Method 2: Binomial theorem / generating function viewpoint

Consider $(x + 1)^{10} = \sum_{k=0}^{10} \binom{10}{k} x^k$.

- (a) The coefficient of x^4 equals $\binom{10}{4}$, counting strings with 4 ones:

$$\boxed{\binom{10}{4} = 210}.$$

- (b) “At most four ones” is $\sum_{k=0}^4 \binom{10}{k}$; evaluating gives 386:

$$\boxed{386}.$$

- (c) “At least four ones” is $\sum_{k=4}^{10} \binom{10}{k}$, which equals $2^{10} - \sum_{k=0}^3 \binom{10}{k} = 848$:

$$\boxed{848}.$$

- (d) “Equal numbers” means $k = 5$, so coefficient of x^5 is $\binom{10}{5} = 252$:

$$\boxed{252}.$$

Question 4 (Ex. 6.3.16)

Problem

Thirteen people on a softball team show up for a game.

- (a) How many ways are there to choose 10 players to take the field?
- (b) Of the 13 people who show up, three are women. How many ways are there to choose 10 players to take the field if at least one of these players must be a woman?

Solution

Method 1: Combinations and complement counting

- (a) Choosing 10 out of 13 without order gives

$$\binom{13}{10} = \binom{13}{3} = 286.$$

286

- (b) Let there be 3 women and 10 men. Total ways to choose 10 players is $\binom{13}{10} = 286$. Subtract the selections with *no* women, i.e. all 10 men:

$$\binom{10}{10} = 1.$$

Hence the number with at least one woman is

$$286 - 1 = 285.$$

285

Method 2: Casework on number of women selected

- (a) A different viewpoint: choosing 10 players is equivalent to choosing 3 players to sit out:

$$\binom{13}{3} = 286.$$

286

- (b) Select exactly r women, where $r \in \{1, 2, 3\}$. Then select $10 - r$ men from 10 men:

$$\sum_{r=1}^3 \binom{3}{r} \binom{10}{10-r}.$$

Compute term-by-term:

$$\binom{3}{1}\binom{10}{9} = 3 \cdot 10 = 30, \quad \binom{3}{2}\binom{10}{8} = 3 \cdot 45 = 135, \quad \binom{3}{3}\binom{10}{7} = 1 \cdot 120 = 120.$$

Sum:

$$30 + 135 + 120 = 285.$$

$$\boxed{285}$$

Question 5 (Ex. 6.3.25)

Problem

How many bit strings contain exactly eight 0's and 10 1's if every 0 must be immediately followed by a 1?

Solution

Method 1: Block method (treat each 0 as part of a mandatory “01” block)

Because every 0 is immediately followed by a 1, each 0 must occur as the first character of a substring “01”. Since there are 8 zeros, we must have exactly 8 blocks

$$B := 01.$$

These 8 blocks consume 8 ones, leaving

$$10 - 8 = 2$$

additional ones that are not forced to follow a 0.

Thus the entire string is formed by arranging:

- 8 identical blocks $B = 01$,
- 2 identical single symbols 1.

There are $8 + 2 = 10$ objects total, with 2 indistinguishable single 1's (equivalently, with 8 indistinguishable blocks). Hence the number of distinct arrangements is

$$\binom{10}{2} = 45,$$

by choosing which 2 of the 10 positions are occupied by the extra single 1's.

$$\boxed{45}$$

Method 2: Stars-and-bars on gaps between mandatory blocks

First line up the 8 mandatory blocks in order:

$$B B B B B B B B.$$

There are 9 gaps where extra 1's can be inserted:

$$_ B _ B _ B _ B _ B _ B _ B _ ,$$

namely: before the first block, between consecutive blocks (7 gaps), and after the last block.

We must distribute the 2 extra identical 1's among these 9 gaps. Let $x_1, \dots, x_9 \in \mathbb{N} \cup \{0\}$ be the number of extra 1's placed in each gap. Then

$$x_1 + x_2 + \dots + x_9 = 2.$$

The number of nonnegative integer solutions is

$$\binom{2+9-1}{9-1} = \binom{10}{8} = \binom{10}{2} = 45.$$

Each solution corresponds bijectively to a unique valid string.

45

Question 6 (Ex. 6.4.2 and Ex. 6.4.5)

Problem

- (a) Find the coefficient of x^5y^8 in $(x+y)^{13}$.
- (b) Give a formula for the coefficient of x^k in the expansion of $(x + \frac{1}{x})^{100}$, where k is an integer.

Solution

Method 1: Binomial theorem (algebraic extraction)

- (a) By the binomial theorem,

$$(x+y)^{13} = \sum_{r=0}^{13} \binom{13}{r} x^r y^{13-r}.$$

To obtain x^5y^8 we need $r = 5$ (since then $13 - r = 8$). Hence the coefficient is

$$\binom{13}{5}.$$

Compute:

$$\binom{13}{5} = \frac{13 \cdot 12 \cdot 11 \cdot 10 \cdot 9}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 1287.$$

1287

- (b) Expand:

$$\left(x + \frac{1}{x}\right)^{100} = \sum_{r=0}^{100} \binom{100}{r} x^r \left(\frac{1}{x}\right)^{100-r} = \sum_{r=0}^{100} \binom{100}{r} x^{r-(100-r)} = \sum_{r=0}^{100} \binom{100}{r} x^{2r-100}.$$

Thus the exponent of x is $2r - 100$. We get a contribution to x^k exactly when

$$2r - 100 = k \iff r = \frac{k + 100}{2}.$$

Therefore:

$$[x^k] \left(x + \frac{1}{x}\right)^{100} = \begin{cases} \binom{100}{\frac{k+100}{2}}, & \text{if } k \text{ is even and } -100 \leq k \leq 100, \\ 0, & \text{otherwise.} \end{cases}$$

$$[x^k] \left(x + \frac{1}{x}\right)^{100} = \begin{cases} \binom{100}{\frac{k+100}{2}}, & k \equiv 0 \pmod{2}, -100 \leq k \leq 100, \\ 0, & \text{otherwise.} \end{cases}$$

Method 2: Combinatorial coefficient interpretation

(a) Write $(x + y)^{13}$ as a product of 13 identical factors:

$$(x + y)^{13} = \underbrace{(x + y)(x + y) \cdots (x + y)}_{13 \text{ factors}}.$$

To obtain a term $x^5 y^8$, we must choose x from exactly 5 of the factors and choose y from the remaining 8 factors. The number of ways to choose which 5 factors contribute x is $\binom{13}{5}$. Each such choice contributes exactly one $x^5 y^8$ term. Hence the coefficient is $\binom{13}{5} = 1287$.

$$\boxed{1287}$$

(b) Similarly,

$$\left(x + \frac{1}{x}\right)^{100} = \prod_{i=1}^{100} \left(x + \frac{1}{x}\right).$$

From each factor we choose either x or x^{-1} . Suppose we choose x from exactly r factors and x^{-1} from the remaining $100 - r$ factors. Then the resulting exponent is

$$x^r \cdot x^{-(100-r)} = x^{2r-100}.$$

To obtain x^k we need $2r - 100 = k$, i.e. $r = (k + 100)/2$. The number of ways to choose which r factors contribute x is $\binom{100}{r}$. This yields the same coefficient formula:

$$[x^k] \left(x + \frac{1}{x}\right)^{100} = \begin{cases} \binom{100}{\frac{k+100}{2}}, & k \equiv 0 \pmod{2}, -100 \leq k \leq 100, \\ 0, & \text{otherwise.} \end{cases}$$

Question 7 (Ex. 8.5.16)

Problem

How many elements are in the union of four sets if each set has 100 elements, each pair of the sets shares 50 elements, each three of the sets share 25 elements, and there are 5 elements in all four sets?

Solution

Let the sets be A_1, A_2, A_3, A_4 .

Method 1: Inclusion–exclusion principle

The inclusion–exclusion formula for four sets is:

$$|A_1 \cup A_2 \cup A_3 \cup A_4| = \sum_{i=1}^4 |A_i| - \sum_{1 \leq i < j \leq 4} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq 4} |A_i \cap A_j \cap A_k| - |A_1 \cap A_2 \cap A_3 \cap A_4|.$$

Plug in the data:

- $\sum_{i=1}^4 |A_i| = 4 \cdot 100 = 400$;
- there are $\binom{4}{2} = 6$ pairs, each with intersection size 50, so the pair-sum is $6 \cdot 50 = 300$;
- there are $\binom{4}{3} = 4$ triples, each with intersection size 25, so the triple-sum is $4 \cdot 25 = 100$;
- the 4-way intersection has size 5.

Therefore

$$|A_1 \cup A_2 \cup A_3 \cup A_4| = 400 - 300 + 100 - 5 = 195.$$

195

Method 2: Count elements by membership multiplicity (exactly r sets)

Let x_r be the number of elements that belong to exactly r of the four sets, for $r = 1, 2, 3, 4$. Then

$$|A_1 \cup A_2 \cup A_3 \cup A_4| = x_1 + x_2 + x_3 + x_4.$$

We are given $x_4 = 5$.

Now write equations from the provided intersection totals.

(i) Sum of set sizes. Each element counted in exactly r sets contributes r to the total $\sum |A_i|$. Hence

$$\sum_{i=1}^4 |A_i| = x_1 + 2x_2 + 3x_3 + 4x_4.$$

Given $\sum |A_i| = 400$ and $x_4 = 5$,

$$x_1 + 2x_2 + 3x_3 + 20 = 400 \iff x_1 + 2x_2 + 3x_3 = 380.$$

(ii) Sum of pairwise intersections. An element in exactly r sets lies in exactly $\binom{r}{2}$ distinct pairwise intersections, so

$$\sum_{1 \leq i < j \leq 4} |A_i \cap A_j| = \binom{2}{2}x_2 + \binom{3}{2}x_3 + \binom{4}{2}x_4 = x_2 + 3x_3 + 6x_4.$$

Given the left side is $6 \cdot 50 = 300$ and $x_4 = 5$,

$$x_2 + 3x_3 + 30 = 300 \iff x_2 + 3x_3 = 270.$$

(iii) Sum of triple intersections. An element in exactly r sets lies in exactly $\binom{r}{3}$ triple intersections, so

$$\sum_{1 \leq i < j < k \leq 4} |A_i \cap A_j \cap A_k| = \binom{3}{3}x_3 + \binom{4}{3}x_4 = x_3 + 4x_4.$$

Given the left side is $4 \cdot 25 = 100$ and $x_4 = 5$,

$$x_3 + 20 = 100 \iff x_3 = 80.$$

Then from $x_2 + 3x_3 = 270$ we get

$$x_2 + 3 \cdot 80 = 270 \iff x_2 = 30.$$

Finally from $x_1 + 2x_2 + 3x_3 = 380$ we get

$$x_1 + 2 \cdot 30 + 3 \cdot 80 = 380 \iff x_1 + 60 + 240 = 380 \iff x_1 = 80.$$

Therefore

$$|A_1 \cup A_2 \cup A_3 \cup A_4| = x_1 + x_2 + x_3 + x_4 = 80 + 30 + 80 + 5 = 195.$$

Question 8 (Ex. 6.4.13)

Problem

Prove that if n and k are integers with $1 \leq k \leq n$, then

$$k \binom{n}{k} = n \binom{n-1}{k-1}.$$

- (a) Give a combinatorial proof. *Hint: Show that the two sides count the number of ways to select a subset with k elements from a set with n elements and then an element of this subset.*
- (b) Give an algebraic proof based on the factorial formula for $\binom{n}{r}$.

Solution

Method 1: Combinatorial double counting

Let X be an n -element set, say $X = \{1, 2, \dots, n\}$. We count the number of ordered pairs (S, ℓ) where:

- $S \subseteq X$ is a subset with $|S| = k$ (a committee of size k),
- $\ell \in S$ is a distinguished element of S (a chosen leader from the committee).

Way 1 (choose S then choose ℓ). Choose the k -subset S in $\binom{n}{k}$ ways, then choose ℓ from S in k ways. Total:

$$\binom{n}{k} \cdot k.$$

Way 2 (choose ℓ then choose the rest of S). First choose ℓ from X in n ways. After fixing ℓ , we must choose the remaining $k-1$ elements of S from $X \setminus \{\ell\}$, which has $n-1$ elements. That can be done in $\binom{n-1}{k-1}$ ways. Total:

$$n \binom{n-1}{k-1}.$$

Both expressions count the same set of ordered pairs (S, ℓ) , hence they are equal:

$$k \binom{n}{k} = n \binom{n-1}{k-1}.$$

$$\boxed{k \binom{n}{k} = n \binom{n-1}{k-1}}$$

Method 2: Algebraic proof using factorial definitions

Using $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, compute the left-hand side:

$$k \binom{n}{k} = k \cdot \frac{n!}{k!(n-k)!} = \frac{n!}{(k-1)!(n-k)!},$$

since $k/k! = 1/(k-1)!$.

Now compute the right-hand side:

$$n \binom{n-1}{k-1} = n \cdot \frac{(n-1)!}{(k-1)!((n-1)-(k-1))!} = n \cdot \frac{(n-1)!}{(k-1)!(n-k)!} = \frac{n!}{(k-1)!(n-k)!}.$$

Thus both sides are identical:

$$k \binom{n}{k} = n \binom{n-1}{k-1}.$$

$$\boxed{k \binom{n}{k} = n \binom{n-1}{k-1}}$$

Question 9 (Ex. 6.4.20)

Problem

Show that if n is a positive integer, then

$$\binom{2n}{2} = 2\binom{n}{2} + n^2.$$

- (a) by algebraic manipulation;
- (b) using a combinatorial argument.

Solution

Method 1: Algebraic manipulation (as requested in part (a))

We compute each side in closed form.

Left-hand side:

$$\binom{2n}{2} = \frac{(2n)(2n-1)}{2} = n(2n-1).$$

Right-hand side:

$$2\binom{n}{2} + n^2 = 2 \cdot \frac{n(n-1)}{2} + n^2 = n(n-1) + n^2 = n(2n-1).$$

Since both sides equal $n(2n-1)$, the identity holds.

$$\boxed{\binom{2n}{2} = 2\binom{n}{2} + n^2}$$

Method 2: Combinatorial proof (as requested in part (b))

Consider a set X of $2n$ people split into two groups of size n , say

$$X = G_1 \sqcup G_2, \quad |G_1| = |G_2| = n.$$

We count the number of ways to choose a *pair* of people from X .

Way 1. Choose any 2 from $2n$: $\binom{2n}{2}$.

Way 2. Break into disjoint cases:

- both chosen from G_1 : $\binom{n}{2}$ ways;
- both chosen from G_2 : $\binom{n}{2}$ ways;
- one chosen from G_1 and one from G_2 : $n \cdot n = n^2$ ways.

Summing gives

$$\binom{n}{2} + \binom{n}{2} + n^2 = 2\binom{n}{2} + n^2.$$

Equating Way 1 and Way 2 yields the identity.

$$\boxed{\binom{2n}{2} = 2\binom{n}{2} + n^2}$$

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Question 10 (Ex. 6.4.21)

Problem

Give a combinatorial proof that

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}.$$

Hint: Count in two ways the number of ways to select a committee and to then select a leader of the committee.

Solution

Method 1: Combinatorial double counting (hinted method)

Let X be an n -element set (of people). Count the number of ordered pairs (S, ℓ) where:

- $S \subseteq X$ is a *nonempty* committee,
- $\ell \in S$ is the leader of the committee.

Way 1 (sum over committee size). If $|S| = k$, then there are $\binom{n}{k}$ ways to choose S , and then k ways to choose $\ell \in S$. Summing over $k = 1, 2, \dots, n$ gives

$$\sum_{k=1}^n k \binom{n}{k}.$$

Way 2 (choose leader first). Choose the leader ℓ first: n choices. After fixing ℓ , each of the remaining $n - 1$ people can either be included in S or not, independently. Hence there are 2^{n-1} choices for the rest of the committee. Total:

$$n \cdot 2^{n-1}.$$

Since both ways count the same set of ordered pairs (S, ℓ) , we have

$$\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}.$$

$$\boxed{\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}}$$

Method 2: Algebraic verification via differentiation

Start from the binomial theorem:

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k.$$

Differentiate both sides:

$$n(1+x)^{n-1} = \sum_{k=1}^n k \binom{n}{k} x^{k-1}.$$

Now set $x = 1$:

$$n(1+1)^{n-1} = \sum_{k=1}^n k \binom{n}{k} 1^{k-1} \implies n2^{n-1} = \sum_{k=1}^n k \binom{n}{k}.$$

$$\boxed{\sum_{k=1}^n k \binom{n}{k} = n2^{n-1}}$$

Question 11 (Optional: Additional Questions on Poker Cards)

Problem

A standard deck has 52 cards: 13 face values (2, 3, 4, 5, 6, 7, 8, 9, 10, J , Q , K , A) and 4 suits (clubs, diamonds, hearts, spades). A five-card poker hand is a *set* of five different cards (order does not matter).

- (a) How many four-of-a-kind hands are there?
- (b) How many full house hands are there?
- (c) How many flush hands are there? (Include straight flushes and royal flushes in the flush count.)
- (d) How many two-pair hands are there?

Solution

Method 1: Direct construction (choose ranks, then suits)

- (a) **Four-of-a-kind:** choose the rank of the quadruple (13 choices). For that rank, the four suits are forced (exactly 1 way). The fifth card (the “kicker”) can be any card of a different rank: choose its rank (12 choices) and its suit (4 choices), for $12 \cdot 4 = 48$ choices. Hence

$$13 \cdot 48 = 624.$$

624

- (b) **Full house:** choose the rank of the triple (13 choices), then choose which 3 suits out of 4 for that rank:

$$\binom{4}{3} = 4.$$

Choose the rank of the pair from the remaining 12 ranks, and then choose 2 suits out of 4:

$$\binom{4}{2} = 6.$$

Thus

$$13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2} = 13 \cdot 4 \cdot 12 \cdot 6 = 3744.$$

3744

- (c) **Flush:** choose the suit (4 choices), then choose any 5 cards from the 13 cards of that suit:

$$\binom{13}{5} = 1287.$$

Total:

$$4 \binom{13}{5} = 4 \cdot 1287 = 5148.$$

5148

(d) **Two-pair:** choose the two ranks that will form the pairs:

$$\binom{13}{2}.$$

For each chosen rank, choose 2 suits out of 4, giving $\binom{4}{2}^2 = 6^2 = 36$ ways. Choose the rank of the fifth card from the remaining 11 ranks and its suit (4 choices), giving $11 \cdot 4 = 44$ ways. Therefore,

$$\binom{13}{2} \binom{4}{2}^2 \cdot (11 \cdot 4) = 78 \cdot 36 \cdot 44 = 123552.$$

123552

Method 2: Rank-multiplicity patterns (partition-of-5 viewpoint)

A five-card hand can be described by the *multiplicity pattern* of its ranks. We count each requested hand type by:

(choose ranks consistent with the pattern) $\times$ (choose suits consistent with multiplicities).

(a) **Pattern (4, 1) (four-of-a-kind).** Choose the rank occurring 4 times: 13 ways. The 4 suits are forced: 1 way. Choose the singleton rank: 12 ways, and its suit: 4 ways. Hence $13 \cdot 12 \cdot 4 = 624$.

624

(b) **Pattern (3, 2) (full house).** Choose the triple rank: 13 ways; choose its suits: $\binom{4}{3}$ ways. Choose the pair rank: 12 ways; choose its suits: $\binom{4}{2}$ ways. Thus $13 \binom{4}{3} \cdot 12 \binom{4}{2} = 3744$.

3744

(c) **Pattern “all in one suit” (flush).** First choose the suit: 4 ways. Then choose the set of 5 ranks (since suits are fixed by the suit choice): $\binom{13}{5}$ ways. Thus $4 \binom{13}{5} = 5148$.

5148

(d) **Pattern (2, 2, 1) (two-pair).** Choose the two ranks to be doubled: $\binom{13}{2}$ ways. For each doubled rank, choose 2 suits: $\binom{4}{2}^2$ ways. Choose the singleton rank from the remaining 11 ranks and choose its suit: $11 \cdot 4$ ways. Therefore the total is $\binom{13}{2} \binom{4}{2}^2 (11 \cdot 4) = 123552$.

123552